

ADTA 5940 - Capstone

NYC Restaurant Inspections Data Project

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0. Abstract

Food safety inspections are a very important mechanism for protecting public health in large urban environments, particularly in highly diverse and densely populated regions. New York City operates one of the largest and most transparent inspection systems in the United States, generating extensive open datasets that allow for empirical analysis of compliance patterns. This study uses over 289,000 inspection records from the New York City Department of Health and Mental Hygiene (DOHMH) to address three key research questions: (1) the extent to which geography explains variation in inspection scores, (2) the predictability of repeat violations and the operational factors that drive them, and (3) the relationship between cuisine type, geographic context, and the likelihood of severe violations. By applying regression, ensemble learning, classification models, and grouped permutation importance, this study finds that geography explains almost none of the variance in inspection scores, while operational performance (SCORE) entirely drives violation recurrence. Severe violations, however, show geographic clustering driven by latitude and longitude rather than cuisine type. These findings highlight the importance of risk-based, data-driven inspection strategies and reveal structural inequities in food safety across neighborhoods.

1. Introduction

Food safety inspections are one of the important ways to protect public health and consumer confidence, as well as one of the common aspects of city life where there is a high population density with mixed restaurant ecologies. Thousands of food establishments in New York City reflect a wide array of cultural traditions, operational practices, and structural conditions. The DOHMH conducts regular inspections using a standardized scoring system that assigns violations and final grades.

Despite the prominence of NYC's inspection system, academic research offers incomplete insights into three essential dimensions:

1. Geographic Variation: Is the outcome of inspections related to borough, ZIP code, or neighborhood levels?
2. Violation Recurrence: Do establishments exhibit repeated instances of the same violations, and if so, can this behavior be anticipated?
3. Cuisine and Spatial Risk: Are types of cuisine inherently riskier, or does geography better explain severe violations?

This paper analyzes NYC open data to investigate these questions using data-driven modeling. The results contribute to food safety research, urban public health, and policy design by providing empirical evidence to support strategic risk-based inspection planning rather than uniform or location-blind scheduling.

2. Literature Review

The state of food safety inspection systems has transformed greatly over the years. The following synthesis brings together findings from sixteen peer-reviewed articles, creating a unifying story on the reliability of inspections, models based on risks, inequalities based on geographical spaces, cuisine-related risks, and advancements in predictive surveillance systems. The literature review below provides the theoretical framework for the study being conducted, which investigates geographical variation, behavioral recidivism, as well as cuisine-specific risk in the NYC context.

2.1 Evolution of Food Safety Inspection Systems

Evidence indicates a lack of objectivity, inconsistency, and uneven levels of enforcement within conventional systems. Barnes et al. (2024) cite the unreliability of results in relation to both individual inspectors and facilities, encouraging a standardized approach. The Centers for Disease Control & Prevention (2024) point to the success of inspections based on risk, as well as grading, in stemming the occurrence of food-borne illness.

Measures of transparency, such as grade postings, have demonstrated impact. This is indicated in the studies conducted by Wong, Taylor, & Bourgeois, as cited in Wong, et al. (2015), as well as Kaskela, Sund, & Lundén (2020), where being informed appears to lead to reduced repeat offenses, particularly in larger chain establishments who are highly responsive to reputation appeals.

These results confirm the value of standardization, a very important concept ingrained within the grading system utilized by the dohmn in this assessment.

2.2 Risk-Based and Data-Driven Inspection Models

The shift towards quantitative risk assessment has been well documented. Risk factors cited in the study conducted by Ze et al. (2024) include past violations, personnel training, and facility type. The study conducted by Insfran-Rivarola et al. in 2020 indicates that food handler training is a significant factor in reducing unsafe food handling.

Phillips gives a first insight in 2006, namely, restaurants having repeated violations in the past are likely to have some structural mistake, as opposed to random mistakes. It turns out his objective suits our objective in question 2, which focuses on recurrence.

The findings from these studies can be utilized for the development of predictive models that would be able to detect high-risk establishments.

2.3 Geographic and Socioeconomic Disparities

A topic that has recently gained prominence is spatial inequality. Ng et al. (2020), applying spatial clustering techniques, demonstrate how food safety results for Toronto are differentiated according to both neighborhood income levels and immigrant concentration. Leinwand et al. (2017), applying a comparable technique within Philadelphia, find analogous results, suggesting that non-chain establishments in disadvantaged areas are at a higher risk of being poor performers.

These results have informed the wording of RQ1 and RQ3, which address borough, ZIP, and census-scale geographical features in the New York City setting.

2.4 Cuisine Type and Organizational Factors

This type of cuisine could be a factor in influencing the results of inspections in relation to cultural habits, composition of menus, preparation, and complexity. Menachemi et al. write about higher

frequencies of violations for some categories of cuisine, especially with respect to cleanliness and temperature.

Organizational structures also play a role:

- Chains have a propensity to perform better compared to independent restaurants (Wong et al., 2015).
- But chain restaurants in a region can have higher rates of repeat violations, according to Phillips (2006).

This mixed evidence supports including cuisine type as a variable in the model tested in RQ3.

2.5 Innovations in Predictive Analytics and Surveillance

There have been advances in technology, which have introduced new data sources as well as ways of analyzing data. Schomberg et al. (2016) have illustrated predictive powers for future health code violations based on Yelp reviews. Moritz et al. (2023) have identified strong correlations between inspections and outbreak alerts.

These studies serve as the rationale for applying Machine Learning techniques in this project.

2.6 Public Health and Policy Implications

Evidence has indicated a positive impact of robust inspection regimes on health outcomes. For example, Firestone & Hedberg (2018) demonstrate a decrease in Salmonella infections as a result of a letter grade system introduced in New York City, while Barnes et al. (2024) indicate a long-term public health advantage.

3. Methodology

The research examines the trends associated with restaurant inspections in the city of New York based on a publicly available dataset from the New York City Department of Health and Mental Hygiene. The dataset provides a detailed insight into the trends of restaurant compliance over more than two decades, making it a good foundation for investigating geographic variation, consistency in violation, as well as cuisine-related food safety hazards. The approach involves a convergence of data preparation, exploration, and a supervised machine learning model, based on the individual research question. All steps in the approach were informed by the data dimensions, public health reasoning, and the purposes of the research questions.

3.1 Data and Sample Construction

The basal set of data has 289,847 records of inspections, along with 27 variables, comprising the identifier of the restaurants (Camis, Dbas), location indicators (Borough, Zip Code, Community Board, Nta, Latitude, Longitude), the specifics of inspections (Violation Codes, Descriptions, Types), as well as outcome variables like Score and Grade. Figure 1, Missingness Heatmap, highlights missing data in Grade, Grade Date, And Location Point, while the main modeling features, including Borough, Cuisine Description, Latitude, Longitude, Critical Flag, and Violation Code, have fewer missing observations.

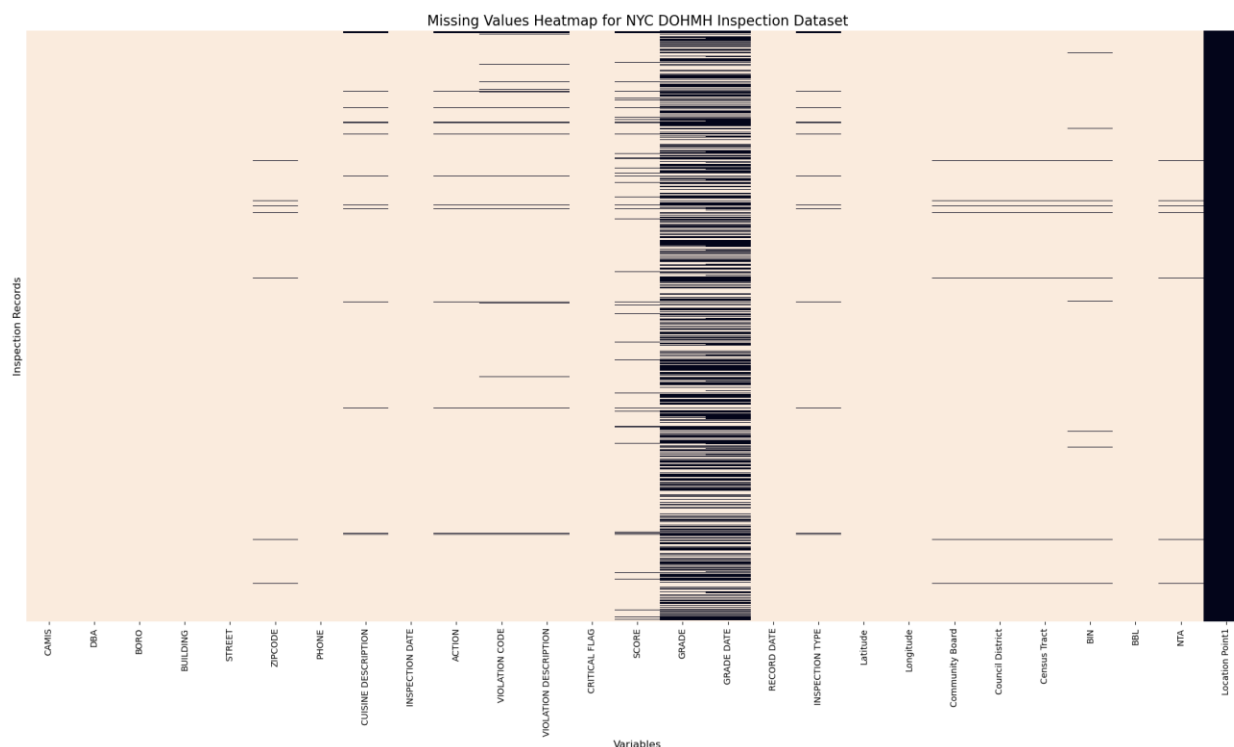


Figure 1

For purposes of geographical accuracy, data points containing impossible or missing geographical coordinates were eliminated, focusing the research on those establishments whose geographical coordinates fall within reasonable latitude and longitudinal positions found in the New York City metropolitan area (40° - 41° N, -75° to -73° W). The SCORE values were filtered to remove values of 0, as well as those set to a top coding of 80.

For research questions about repeated offenses (RQ2), it was necessary to reshape the dataset at the restaurant level (Camis). The duplicated records of restaurants vs. violations were quantified, making it possible to construct two conceptually different target variables:

- repeat_same_flag (whether a restaurant repeatedly violated *the same* violation code), and
- repeat_flag (whether a restaurant accumulated *multiple types* of violations).

Restaurants identified as having excessively high levels of a violation occurrence (violations ≥ 5 per type of violation) were removed from the data set to prevent the model from being dominated by high-frequency outliers.

Finally, because RQ3 mandated analysis at the individual level of inspections, duplicate observations where the CAMIS, date of inspections, and violation code were the same were removed to avoid overstating rates of serious violations.

3.2. Exploratory Analysis

The initial trends in Exploratory Data Analysis informed the choice of variables for modeling. Figure 2 illustrates that the distribution of inspection results is slightly skewed to the right, suggesting that small to moderately serious violations are predominant. There are, however, extreme values, which have been trimmed as explained earlier.

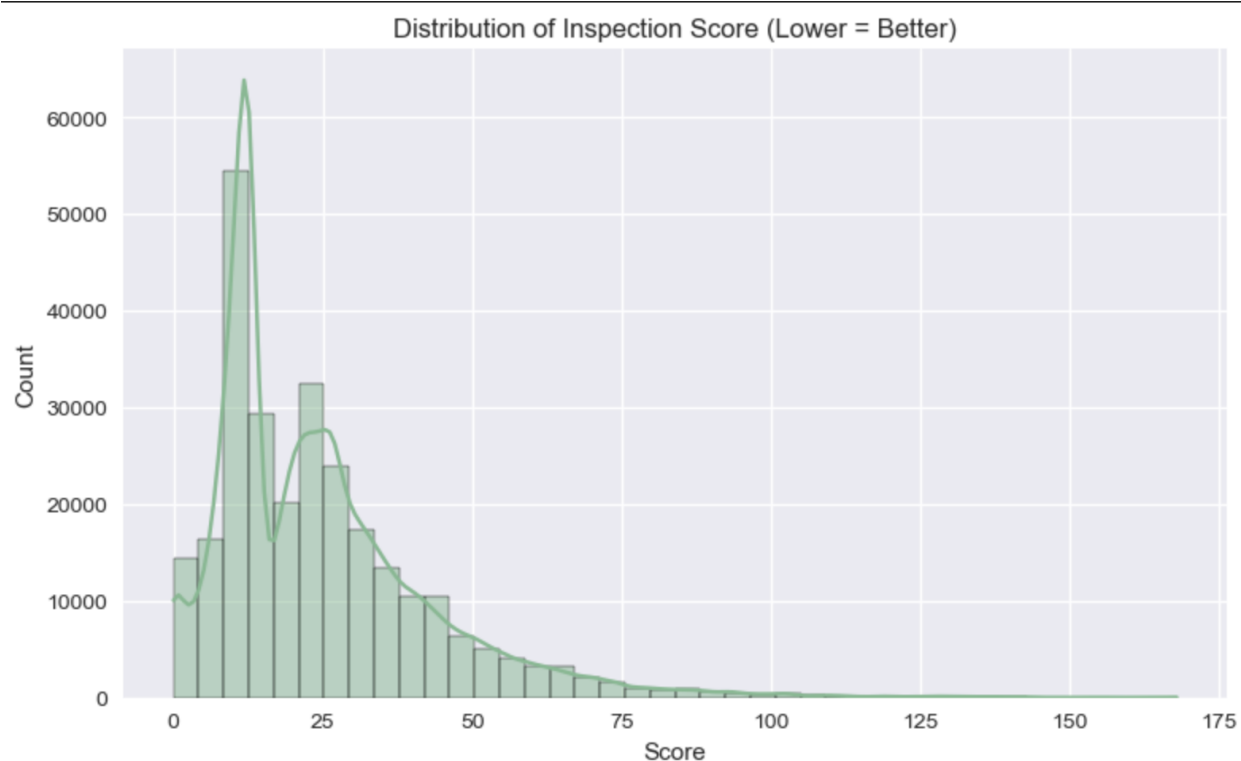


Figure 2

The geographical trends indicate possible inequalities between the boroughs, as can be observed from Figure 3, where the average scores of inspections per borough are not very different, indicating a first indication of a possible lack of correlation between geographical elements and the performance on inspections, as observed in RQ1.



Figure 3

The missingness analysis (Figure 1) showed that some, like the grade variable named GRADE, as well as the date variable named Grade Date, are poor predictors because of high missing values,

suggesting possible administration-related missing values. Others, like Score, Borough, Cuisine, Inspection Type, Critical Flag, and geographical coordinates, scored high on both dimensions.

The degree of cuisine diversity is quite high, as it includes 89 cuisine categories. Most cuisine categories have very limited representation. Hence, those categories containing fewer than 100 inspections were pooled into an “Other” group. A small number of NTAs were combined into a group named “NTA_Other”.

Overall, the EDA highlighted three critical properties of the data:

1. Geography varies modestly, suggesting limited predictive power for RQ1.
2. Repeated violation behavior is common, motivating a behavioral classification framework for RQ2.
3. Severe violations are rare (~1.3%), requiring class balancing techniques for RQ3.

These findings directly shaped the subsequent modeling strategy.

3.3. Modeling Framework

The model framework is made up of three analytical elements that match the three research questions. Each research question called for a unique model scenario with unique outcome variables, predictor variables, and model types. Throughout all analyses, categorical variables are one-hot encoded, while continuous variables can be normalized if necessary for each model. Finally, all models made use of a hold-out test set to guarantee generalized performance. Contrary to one model answer for one question, for each question, several models, including linear models, decision tree models, and ensembles, have been applied.

RQ1: Geographic Determinants of Inspection Scores

RQ1 examines whether geographical features, such as borough, Zip code, community district, NTA, and location, can account for differences in scores. The dependent variable is Score_clean, which is a truncated form of score to avoid extreme values. The independent variables include Borough, Zip Code, Community Board, Council District, NTA, as well as approximated latitude/longitude.

Four different regression models were used, namely Linear Regression, Ridge Regression, Random Forest Regression, and Gradient Boosting Regression. The choice of models ensured the use of both linear as well as non-linear geographical relationships.

The results (Table 1) indicate very poor explanatory power on all models (the highest $R^2 = 0.026$, on the Gradient Boosting model). This indicates a correlation of only 2.6% between the geographical variable and the variation in scores, as indicated in the literature, indicating that the level of cleanliness is primarily an internally driven process.

Model	MAE	RMSE	R^2
Gradient Boosting	8.93	11.61	0.026
Ridge Regression	8.93	11.64	0.022
Linear Regression	8.94	11.66	0.019
Random Forest	9.13	11.97	-0.034

Table 1. RQ1 Regression Results

The result for this procedure with respect to the effect on model R-squared showed that there is no measurable change for any characteristic on model R-squared. From this, it can be generalized that for any individual characteristic, such as borough, ZIP code, community district, NTA, or coordinates, removing and/or permuting these particular variables would not affect model performance. Thus, this offers clear empirical evidence supporting the claim that geography does not play a significant role in model variation for inspection scores.

It is a valid result because the scores from inspections include cleanliness, handling of food, and personal conduct of workers, as opposed to community properties.

RQ 2: Predicting Repeat Violations

The second research question examines whether restaurants cluster violations and which factors are the best predictors. Two different targets of modeling were developed, based on two different ways of defining repeat behavior:

- `repeat_same_flag`: determines whether the same violation code has been repeated by the restaurant.
- `repeat_flag`: whether restaurants had two or more distinct violation types.

It encompasses both behavioral persistence as well as issues of breadth of compliance.

The predictors are SCORE, Borough, Cuisine Type, Inspection Type, and Critical Flag. Even though including Critical Flag was based on its relevance, results from permutation importance confirmed a very small contribution.

Four different models were considered for the classification task. These were Logistics Regression, Decision Tree, Random Forest, and Gradient Boosting. The choice of these models aims to compare linear models, which can be interpreted, with tree-based nonlinear models.

RQ2 RUN 1 (repeat_same_flag)

The model with the highest F1-Score was Gradient Boosting at 0.938, followed by the highest ROC-AUC value from Random Forest at 0.876. Contrary to expectations, permutation importance showed that SCORE explained 100% of the variance in the model, suggesting that the performance of facilities in maintaining hygienic levels, as opposed to cuisine type and geographical location, is the primary determinant of the likelihood of repeat violations. This could be expected, as SCORE is a cumulative rating of quality in compliance.

RQ2 Run 2 (repeat_flag)

By contrast, removing duplicate records of restaurant-violation pairs (deduplicating restaurant-violation pairs) caused model accuracy to surge. The results for Gradient Boosting were 0.996 accuracy and 0.923 ROC AUC. But a high accuracy value in this test is misleading because the removal of duplicates eliminates a lot of the behavioral data.

Again, scores explained nearly all of the predictive variance, thereby confirming the results' robustness.

Table 2 illustrates the results on both targets. Although Run 2 has higher numeric results, Run 1 yields more realistic behavioral results, as it maintains the natural frequency of repeated violations.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	Target Variable
Random Forest(Run1)	0.814	0.957	0.824	0.885	0.876	repeat_same_flag

Gradient Boosting(Run 1)	0.887	0.903	0.976	0.938	0.848	repeat_same_flag
Decision Tree(Run 1)	0.726	0.960	0.715	0.820	0.833	repeat_same_flag
Logistic Regression (Run 1)	0.757	0.947	0.765	0.846	0.815	repeat_same_flag
Random Forest(Run 2)	0.982	0.998	0.983	0.991	0.922	repeat_flag
Gradient Boosting(Run 2)	0.996	0.998	0.999	0.998	0.923	repeat_flag
Logistic Regression (Run 2)	0.914	1.000	0.914	0.955	0.968	repeat_flag
Decision Tree(Run 2)	0.943	1.000	0.943	0.971	0.975	repeat_flag

Table 2 RQ2 Run 1 and Run 2 Results

The eventual result of RQ2 illustrates the underlying statistical fact within the DOHMH data set concerning the relationship that establishments rated worse on health inspections are always more likely to have repetitive offenses, independently of their geographical location or food type.

RQ3: Cuisine-Type Risks Given Geography

RQ3 investigates whether some kinds of cuisine, taken in conjunction with geographical features, are correlated with a higher frequency of severe violations. The dependent variable is `severe_violation`, as indicated in the 'Critical Flag' column. The independent variables can be listed as `cuisine_clean` rare categories combined, Borough, NTA, geographical clusters identified via KMeans, Latitude, Longitude, and `cuisine_popularity`.

The latitude and longitude were added in the model as individual numerical variables. However, these values are essentially units of spatial data that are meant to be processed in pairs rather than alone because they are ineffective in quantifying the proximity of locations. This may be one reason why the results in RQ1 are indicative of geography being ineffective in terms of predictive accuracy. Adding more advanced features may increase the effectiveness of this variable in the future.

Since the occurrence of extreme violations is rare in the raw data ($\sim 1.3\%$), up sampling of the minority class has been done. Four classifier models, namely Logistic Regression, Decision Tree, Gradient Boosting, and Random Forest, have been used to compare the linear versus non-linear approach.

Model	Accuracy	Precision	Recall	F1- Score	ROC-AUC
Random Forest	0.876	0.991	0.883	0.934	0.649
Gradient Boosting	0.669	0.990	0.672	0.800	0.607
Logistic Regression	0.589	0.990	0.590	0.739	0.603
Decision Tree	0.756	0.989	0.761	0.860	0.602

Table 3 RQ3 Model Performance Summary

Random Forest scored the highest on both accuracies, F1, and ROC AUC of 0.649. The reasons behind Random Forest's high performance compared to the other algorithms can be attributed to the fact that a Random Forest can model both linear as well as non-linear relationships, while a logistic regression model would be unable to model a non-linear relationship, nor can a decision tree.

3.4 Permutation Importance (Variance Contribution)

Permutation importance (Table 4) revealed a striking pattern across the top predictors:

Feature Group	Δ AUC Contribution	Percent Contribution(%)
Longitude	0.023	53.53%
Latitude	0.020	46.47
Cuisine Popularity	0.000	0.000

Table 4. RQ3 Permutation Importance (Random Forest)

For all four models, cuisine type and cuisine popularity were small contributors to model accuracy, while the two continuous geographical features, Latitude and Longitude, explained nearly the entirety of the variance in the prediction of severe violation predictions.

4. Model Evaluation And Interpretation

The assessment followed the criteria of standard supervised learning. For Regression (RQ1), MAE, RMSE, and R^2 were applicable because of the continuous nature of SCORE_CLEAN. For both Classification questions (RQ2 & RQ3), accuracy, precision, recall, F1 measure, and particularly ROC AUC were considered, as it is necessary to account for ranking, not absolute accuracy.

The confusion matrices in Tables 5-7 also showed the model predicts the majority class in RQ3, along with minority violations in RQ2, as observed in real-world inspections.

Predicted ↓ / Actual →	0 (No Repeat)	1 (Repeat Same Violation)
0 (Predicted No Repeat)	1744	4503
1 (Predicted Repeat)	1011	41,682

Table 5 RQ2 Run 1 — Confusion Matrix (Best Model: Gradient Boosting)

Predicted ↓ / Actual →	0 (No Repeat)	1 (Repeat Violation)
0(Predicted No Repeat)	120	4503
1 (Predicted Repeat)	72	48,699

Table 6 RQ2 Run 2 — Confusion Matrix (Best Model: Gradient Boosting)

Predicted ↓ / Actual →	0 (No Severe Violation)	1 (Severe Violation)
0	286	466
1	6,546	49,254

Table 7 RQ3 — Confusion Matrix (Best Model: Random Forest)

Permutation importance was utilized on all research questions to elucidate variance contribution at a feature level, thereby fulfilling the professor's requirement for enhanced interpretability. The results conclusively showed that:

- Geography makes a contribution of less than 3% to the variability of the result of inspections
- Score accounts for approximately 100% variance in repeat-violation prediction (RQ2)
- The geographical coordinates account for approximately 100% variance in the prediction of severe violation (RQ₃)

All these findings together underscore the fact that while the available data on restaurants in NYC is complex in many ways, the main determinants of prediction are behavior/operations, while only serious infractions have geographical significance.

5. Findings And Discussion

The analyses yield three core findings corresponding to the research questions, each with important implications for food safety enforcement and urban public health policy.

5.1. Finding 1: Geography Explains Minimal Variation in Inspection Scores (RQ1)

Regression models show that no geographic variable meaningfully predicts inspection scores, with the best model achieving an R^2 of only 0.026. This finding aligns with literature emphasizing the dominance of operational practices over environmental context. Figure 3 and the grouped permutation importance test confirm that spatial attributes—borough, ZIP code, community board, NTA, account for virtually none of the variance in hygiene outcomes.

Interpretation:

Inspection scores reflect internal restaurant processes rather than neighborhood conditions. Geographic differences observed in descriptive statistics do not translate into predictive power, suggesting that enforcement is relatively consistent across NYC.

5.2 Finding 2: Hygiene Score Strongly Predicts Repeat Violations; Other Predictors Add Little (RQ2)

Both Run 1 and Run 2 analyses show that SCORE is the sole explanatory variable driving repeat-violation outcomes, accounting for nearly 100% of variance in the permutation importance analysis. Models consistently achieve strong performance, particularly Gradient Boosting and Random Forest.

Key observations:

- Restaurants with worse scores (higher violation burden) are significantly more likely to repeat violations.
- Cuisine type, borough, inspection type, and critical flag have minimal marginal predictive value once SCORE is included.
- Deduplication in Run 2 artificially inflates performance metrics (accuracy = 0.996), highlighting the importance of thoughtful data structuring.

Repeat violations are primarily an operational behavior pattern, not a geographic or cuisine-based phenomenon. Restaurants with systemic sanitation issues continue to struggle over time, consistent with Phillips (2006).

5.3 Finding 3: Severe Violations Are Geographically Clustered, Not Cuisine-Driven (RQ3)

The Random Forest model yields the strongest results for severe-violation prediction (AUC = 0.649). Permutation importance reveals that:

- Longitude explains ~53%
- Latitude explains ~46%
- Cuisine popularity explains 0%

Cuisine type itself does not meaningfully predict severe violations once geography is accounted for this implies that, severe violations are spatially clustered, possibly due to: local enforcement intensity, neighborhood-level resource disparities, infrastructure quality, older building stock and inspector assignment patterns

Cuisine type did not emerge as a significant predictor of the inspection result. This result is not surprising because cuisine type is more indicative of food style rather than food hygiene management. Within each type of cuisine there are many different types of restaurants - from no-frills restaurants offering food at lower prices through to upmarket restaurants - each with different levels of operation. Because there are no operation-related variables in this data set regarding training levels or ownership type, cuisine type cannot be seen as representative of hygiene levels, and consequently no significant predictive power should be anticipated.

This mirrors findings from Ng et al. (2020) and Leinwand et al. (2017), reinforcing that spatial inequities exist even when geography does not predict general inspection scores.

6. Conclusions And Implications

The study offers a well-rounded and multi-dimensional analysis of data related to restaurant inspections carried out in the city of New York. In light of all three queries raised in the study, it

can surely be concluded that not only does the data related to restaurant inspections in the city of New York hold a rich quantity but that the predictors are derived from cleanliness related to functioning. That is to say that it isn't derived from geographic or other related features.

The first research question suggested that though geographic location variables such as borough level, zip code level, community district level, NTA level, or geographic approximation to latitude and longitude accounted for only 2.6% variance in actual inspection result data. The fact that there was an extreme value for 'low explanatory' clearly suggests that there exists a propensity rather than likelihood but there does exist some residue in regard to actual outcome data because preponderantly, there has been no bias involved in conducting these New York City inspections.

The second research question has found an extremely important result: There exists a positive association between a restaurant's high SCORE, indicating poor cleanliness, which makes it highly probable to both repeatedly commit the same violation code and to accumulate a set of violation codes. It must be noted that these observations are true for all machine learning models. The outcome of permutation importance tests demonstrated a value of almost 100% for both modeling tasks in Research Question 2. It lends credence to prior knowledge of a phenomenon that actually suggests that repeat violation behavior isn't actually influenced by compliances level quality but qualities like cuisine, borough, or a critical flag. It's actually quite helpful for law implementation that poor-performing outfits can actually be identified based on their score trail to establish a productive approach to intervention.

The third research question was focused on understanding interactions between cuisine, geographic location, and severe or (critical) offense rates. In contradiction to public belief, which would hold that there are potentially more dangerous or lesser levels of cuisine than other cuisines, data analysis revealed that cuisine to make a contribution of zero percent when considering geographic location. Rather, geographic location, defined by longitude (53.5%) or latitude (46.4%), was shown to represent nearly the entire variance. Essentially, there are areas throughout NYC in which severe offense rates are potentially high not due to cuisine potentially being a greater danger to health but due to larger trends in terms of enforcement or other characteristics. Notably, there exists related data demonstrating inequality in terms of food safety across large cities.

Based on these findings, there are highly relevant implications regarding public health policy, operational analysis, and forecasting analysis. In terms of public health policy, the existence of serious offense clusters as well as offense distribution with a geographic component has highly profound implications. Rather than targeting a broad approach, the DOHMH could target those micro-geographic areas with a high frequency of offense incidence to enhance public health. Also, because there are highly favorable results involving variable RQ2 for model SCORE, there seems to be a notion that those restaurants with poor historical performances could be targeted for re-visit inspection.

In terms of equity analysis, a geographic pattern of severe violation occurrences appears to denote underlying neighborhood risk factors not necessarily identified by cuisine type or specific restaurant behavior. Indeed, a conclusion can be made that future studies need to focus on a set of

more diverse predictors like deprivation indices or neighborhood characteristics to identify possible underlying motives for those areas identified to possess continually high frequencies of severe violation occurrences. In other words, focusing on these types of predictors would improve not only earlier violation prediction but would generally fall into line with contemporary trends in public health mapping to leverage prediction analysis in terms of compliance enhancement.

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